

A Verifiable Search Is Not a Learnable Chain-of-Thought

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June 2026

Abstract

It is tempting to assume that any task one can solve with a short program can be taught to a model as its chain-of-thought: write the program’s steps out in words, fine-tune on them, and the model should follow. This paper shows the assumption fails for an identifiable class of procedures, and that the reflexive remedies do not rescue it. The testbed is nine reasoning tasks, each emitted by a deterministic generator, with one convenient property: the public and hidden splits are drawn from the *same* generators, so a held-out slice of training data is a faithful proxy for test accuracy. I reverse-engineer the generators into Python solvers (five reach $\geq 98\%$), render their procedures as chain-of-thought, and distill them into a rank- ≤ 32 LoRA over a 30B (3.5B-active) Nemotron model. Forward-computable tasks install readily: four lookup/arithmetic tasks and an 8-bit boolean-rule task transfer (≥ 0.99 and 0.68). Cryptarithm does not, when one tries to distill its backtracking *search*: it holds at 0.01 – 0.07 across eleven chain-of-thought designs, RL from verifiable rewards, and self-training, even though a search solver answers 71% of its instances. The failure is not a capability gap. The model does the arithmetic correctly on 97 – 100% of lines and ranks the correct cipher into its top eight on 71% of instances. What it cannot do is carry the *search* forward as a left-to-right derivation. Supervised fine-tuning therefore learns the *shape* of a verifiable elimination step while its verdicts become unconditional templates, correct only 16 – 57% of the time (“verdict-as-token”). This search-distillation ceiling holds across four backbones from 3B to 671B and across both fine-tuning and in-context prompting, and a controlled intervention isolates the cause: revealing the cipher key, which turns the derivation forward, lifts the *same* instances from 0.03 to 0.57 . The constraint binds the *search*, not the task: when a procedure’s only solution is search over information-free structure, no faithful forward chain-of-thought exists to imitate. The task nonetheless becomes learnable by *removing the search from the trace*, precomputing its combinatorial core into a finite catalog the model memorizes and reducing the chain-of-thought to recall plus bounded verification; the competition’s 1st-place solution is the existence proof, reaching Private LB 0.92 by memorizing a per-signature candidate catalog rather than teaching the model to search [8]. What distills is memorization and verification, not search. I release the benchmark decomposition, the solvers, the per-line fidelity audits, and the control experiments.

1 Introduction

If you can write a short program that solves every instance of a task, is the task “learnable” as a chain-of-thought? Intuitively the program is a recipe, the recipe can be written out as natural-language steps, and a capable model fine-tuned on those steps should reproduce them. This intuition underlies a large body of recent work on distilling reasoning into smaller models, from rationale

distillation [11] to bootstrapped self-training [31] and reinforcement learning from verifiable rewards [5, 16, 25].

This paper reports a controlled study in which the intuition fails, sharply and reproducibly, for a specific and identifiable class of procedures. I use a benchmark of nine reasoning tasks, each emitted by a deterministic generator (Roman numerals, linear unit conversion, free-fall, monoalphabetic substitution cipher, an 8-bit boolean program, two equation-induction families, and two cryptarithm families). The benchmark has a property that makes it an unusually clean laboratory: the hidden test set is produced by the *same* generators as the public training set, so the two are draws from one distribution and a stratified held-out slice of training data predicts test accuracy directly. I can therefore measure “did the procedure transfer” to a fraction of a point without access to the test labels.

Research questions. I ask: **(RQ1)** does a verifiable solver’s coverage predict whether its procedure is learnable as a small model’s chain-of-thought? **(RQ2)** when a procedure resists every training method I try, what is the binding bottleneck? **(RQ3)** is there a cheap, pre-training test that predicts resistance? In short: *no*, solver coverage does not predict whether the *search* distills (Figure 1); a search over information-free structure admits no faithful forward CoT, so SFT learns only its unfaithful surface form (“verdict-as-token”), architecture-independently (Section 5.2). The binding bottleneck is forward-derivability, isolated by a controlled intervention that lifts the same instances $0.03 \rightarrow 0.57$ (Table 7). The resistance is predictable in advance, via a three-part screening heuristic (Section 7). And it is not a ceiling on the *task*: the practical escape, demonstrated by the competition’s 1st-place solution, is to *memorize* the search’s finite structure rather than distill the search itself (Section 9).

I first reverse-engineer the generators into Python solvers. Five of the nine tasks admit solvers at $\geq 98\%$ accuracy. Cryptarithm admits a backtracking-search solver that answers $\approx 71\%$ of instances (the remainder are information-theoretically ambiguous). I then attempt to install each solver’s procedure into a rank- ≤ 32 LoRA adapter [12] over **Nemotron-3-Nano-30B-A3B**, a hybrid Mamba-2/Mixture-of-Experts model [4, 22], served greedily under a 7680-token budget. Training is SFT on solver-grounded synthetic CoT generated with *different* values from any evaluation instance (zero leakage); when SFT plateaus I escalate to RLVR/GRPO and STaR.

Thesis. A verifiable *search* does not distill into a chain-of-thought. The dividing line is whether the task admits a *faithful forward chain-of-thought*, a left-to-right derivation that reaches the answer. Forward-computable tasks install readily; a task whose only solution is backtracking search over information-free structure has no faithful forward trace, so SFT copies the *shape* of the search but not its logic, across every architecture and scale I tried. The task is not thereby unlearnable: it becomes learnable by *removing the search from the trace*, precomputing its combinatorial core into a finite catalog the model memorizes and reducing the chain-of-thought to recall plus bounded verification (Section 9). What distills is memorization and verification, not search.

Contributions.

- **A controlled solvability–learnability gap.** On a $\text{train} \equiv \text{test}$ benchmark I quantify, per task, the gap between a verifiable solver’s accuracy and the accuracy the fine-tuned model actually reproduces in CoT (Figure 1, Table 1). The gap is near-zero for four tasks and exceeds 0.65 for cryptarithm.
- **A negative result with convergent evidence.** Across eleven CoT-engineering rounds and three RL/self-training escalations, cryptarithm-deduce never exceeds 0.07 (Figure 4). Five

independent measurements, solver ceiling, information floor, base-model behavior, forward-derivation coverage, and per-line execution fidelity, each independently bound the achievable accuracy, and they agree (Table 3).

- **Three named, measured mechanisms.** *Verdict-as-token* (Figure 5): arithmetic is correct on 97–100% of lines but elimination *verdicts* are emitted as unconditional templates (fidelity 16–57%). *Derivation- vs. ranking-blocked*: a budget-unbounded generate-and-verify reframe shows ranking is not the constraint ($\text{hit@8} = 0.71$) while forward derivation is ($1/659$). *Information-free sub-structure*: the hidden map’s $\text{MI}(\text{digit}; \text{glyph}) = 0.021 \text{ nats} \approx 0$.
- **A positive control that confirms the mechanism.** Hand-written bit-manipulation CoT that *teleports* the rule search collapses to ≈ 0.05 ; STaR on the model’s own successful searches lifts $0.526 \rightarrow 0.678$ and removes truncation ($18.6\% \rightarrow 0.2\%$, Figure 3). What transfers is search the model can actually run, not search narrated for it.
- **An architecture control.** Training the identical cryptarithm corpus on four backbones, hybrid Mamba/MoE, two dense Transformers, and an MoE Transformer, spanning 3B–21B, hits the *same* ≤ 0.04 floor (Table 6). The search-distillation ceiling is architecture-general, not backbone- or scale-specific. This refutes my own initial SSM-specific hypothesis.
- **A causal intervention.** Revealing the cipher key on the *same* cryptarithm instances, turning the derivation forward, lifts Nemotron from 0.03 to **0.571** (Table 7). Revealing only half barely helps. Forward-derivability is the causal lever, and a faithful forward CoT must cover the *whole* derivation, not most of it.
- **The escape, externally validated.** The search-distillation ceiling is *not* a ceiling on the task. The competition’s 1st-place solution reaches Private LB 0.92 by *memorizing* the search’s finite structure, a 4,205-entry per-signature candidate catalog for cryptarithm and a 5,238-entry rule-sequence catalog for bit-manipulation, and verifying candidates in-trace, rather than distilling the search [8]. This both confirms the mechanism, why memorization is necessary, and bounds the claim (Section 9).

I did not win the competition this benchmark is drawn from, my best adapter scored 0.85 public (0.86 private, on an unselected submission, Section 8) against a leader at 0.92. As I discuss in Section 8, 0.85 is the *median* of the 4,355-team field and the ceiling reachable without the memorization–computation reformulation of cryptarithm and bit-manipulation (Section 9); a fully open-source solution taking my same search-distillation approach won the competition’s Open Progress Prize at the same 0.85. The contribution is not a ranking; it is a clean, instrumented account of *where and why* a verifiable search fails to become model reasoning, and of what does transfer in its place.

2 Background and Related Work

Chain-of-thought and its distillation. Eliciting step-by-step reasoning improves LLM accuracy on multi-step tasks [29, 30], and a model’s rationales can be distilled into smaller students [10, 11, 21]. My setting is distillation in the strong sense: the “teacher” is a perfect symbolic solver, and I ask whether its procedure survives transfer into a small model’s CoT under a budget.

Self-training and verifiable rewards. STaR bootstraps a model on its own correct rationales [31]; RL from verifiable rewards and group-relative policy optimization (GRPO) optimize directly against an answer checker [5, 16, 25], and process verifiers supervise intermediate steps [18]. The

generate-and-verify idea traces to training answer verifiers on grade-school math [3]. I use all three as escalations. A central finding is that on my hard task RLVR’s gradient is *present but non-transferring*: it improves easy curricula yet does not move held-out accuracy, because positive rollouts on real instances are too rare to learn from.

Faithfulness of chain-of-thought. A growing literature shows that a model’s stated reasoning need not reflect the computation behind its answer: explanations can be systematically unfaithful [28], and interventions on the trace (inserting mistakes, truncating, paraphrasing) often leave the answer unchanged [17], which motivates constructions that force the answer to follow from the stated steps [19]. My *verdict-as-token* result is a sharp, mechanistic instance of unfaithfulness arising in the *training* regime: SFT installs the surface form of a verifiable elimination step while decoupling it from the step’s logic, so the imitated trace is unfaithful by construction at free-running inference.

Limits of transformers on procedures. My negative result is consistent with, and sharpens, work on compositional and algorithmic limits: transformers degrade on multi-step composition as depth grows [7] and struggle to generalize algorithm length [1]. Length generalization holds only for tasks expressible as short length-invariant programs [32], memory-free architectures cannot climb the Chomsky hierarchy [6], algorithmic competence can emerge only after delayed generalization [23], and models score poorly even when worked step-by-step solutions exist [9]. I add a *mechanism* specific to imitation learning of search, verdict-as-token under teacher forcing, and connect it to the classic exposure-bias gap between teacher-forced training and free-running inference [2, 24].

Architecture. The base model is a hybrid in which most attention layers are replaced by Mamba-2 state-space layers, interleaved with a sparse Mixture-of-Experts [4, 22], and served with paged-attention batching [15]. State-space layers compress history into a fixed-size state; I will argue this interacts unfavorably with tasks that require explicit, unbounded search-state. Only 6 of the model’s ~ 52 layers carry attention; the bulk are Mamba-2 or MoE. The rank- ≤ 32 adapter I train spans the attention, Mamba, MoE, and output projections alike (Appendix C), but a rank-32 update is a thin slice of a model whose capacity sits overwhelmingly in layers that summarize history into fixed state. There is theory for why this should bite: state-space layers lie in TC^0 and cannot perform genuine sequential state-tracking [20], and a fixed-size recurrent state caps copying and retrieval from context [13]. I tested this directly, however, and find it is *not* the explanation for cryptarithm (Table 6): dense and MoE-Transformer models of comparable active size hit the same floor, so the SSM limit is at most a contributing factor (e.g. for the bit-manipulation tail), not the cause of the search-distillation ceiling.

3 The Testbed: a Deterministic-Generator Benchmark

Task and grading. The benchmark ships a public training set of 9,500 labeled problems and is scored on a hidden test set of ≈ 500 problems. A submission is a single LoRA adapter of rank at most 32 loaded onto the frozen base model and served with vLLM in deterministic (greedy) mode (temperature 0, top- p 1.0, `max_tokens= 7680`, `max_model_len= 8192`), with thinking enabled (`<think>...</think>`). The model must place its final answer in a `\boxed{}` field; the grader extracts the last such field (falling back to the last number) and applies a fixed `verify()`: exact case-insensitive string match for binary strings, relative tolerance 10^{-2} for floats, exact string match otherwise.

The train $\equiv$ test property. Every category is produced by a deterministic generator, and the hidden test uses the *same* generators. Train and test are therefore i.i.d. draws from one distribution, and a stratified held-out slice of `train.csv` (100 rows/category) is a faithful oracle for the leaderboard. I exploit this throughout: I train on synthetic CoT built with different values (zero leakage) and evaluate on real held-out training rows, treating that number as the test accuracy. A determinism check confirms the regime: when an in-class rule fits a problem’s worked examples, it yields a unique prediction on the query 98.7% of the time and is correct 99.8% of the time, the examples *pin* the rule, so the only limits to 100% are solver and model completeness, not intrinsic ambiguity (cryptarithm, below, is the exception that proves this).

The nine generators. Table 1 lists the categories, their test weights, and the base model’s zero-shot accuracy. Briefly: *numeral* (write an integer in Roman numerals), *unit_conversion* (a linear map $y = ax + b$ pinned by two examples), *gravity* (free-fall $d = \frac{1}{2}gt^2$, a one-parameter fit), and *cipher* (a monoalphabetic substitution over a fixed 77-word vocabulary, solved by aligning example word-pairs and constraint-propagating the rest) are the “easy four.” *bit_manipulation* maps 8 bits to 8 bits by a single global boolean function over up to three *taps* (rotated/shifted/negated copies of the input), the same function for all eight output bits, pinned by 7–10 examples. *equation_numeric* (deduce/guess) shows visible-digit equations $AB[\text{op}]CD = \text{RHS}$ from which one must induce the operand pairing, the operation, and an output-string format. *cryptarithm* (deduce/guess) is the hard case, described next.

Cryptarithm structure. Each example is a five-character left-hand side `s0 s1 OP s2 s3 = RHS`. Three things are hidden: (1) a per-row injective digit $\rightarrow$ symbol cipher (a uniform draw of 10 distinct glyphs from a 23-symbol pool); (2) an *operation* attached to the middle operator glyph, drawn from a wide vocabulary (add, sub, mul, abs-difference, mod, gcd, lcm, floor-div, the $\pm 1/\pm 2$ variants, and two purely structural *concat* operations). And (3) a per-row *endianness* (standard or little-endian, reversing operand and result strings). “Deduce” means the query’s operator was seen in the examples; “guess” means it was not. Because the cipher is information-free, no single column reveals a digit: the map is pinned only by requiring *all* example equations to hold simultaneously under the right operations and direction, a cross-equation constraint search with backtracking. This is the structural reason cryptarithm sits on the far side of the learnability frontier.

Compute and budget. Final adapters are trained on a single RTX PRO 6000 (Blackwell, 96 GB) via the competition platform. The adapter has $\approx 880\text{M}$ trainable parameters (2.71% of the model). Cheap iteration on data and CoT design is done on a managed LoRA service at $\approx \$3.5/\text{round}$ before promoting a finalized dataset to the full training run, a three-tier “cheapest-first” ladder (API probe $\rightarrow$ cheap LoRA $\rightarrow$ full train) that I found essential for keeping a long negative-result search affordable. All CoT is written in plain ASCII: the tokenizer is a 131,072-entry BPE with no `<unk>`, but Unicode operators ($\oplus, \neg, \neq$) fragment into rare multi-byte tokens, so I spell operators as words (XOR, NOT), arrows as `->`, and inequality as `!=`.

4 Method: Solver-Grounded Synthetic CoT and the Experiment Ladder

My pipeline is deliberately simple, so that any failure is attributable to learnability rather than to data noise. For each category I (1) reverse-engineer the generator into a solver and validate it against the 9,500 training rows; (2) *render* a synthetic CoT for fresh, randomly generated instances

Table 1: The benchmark, and the central gap. “Solver” is the accuracy of my reverse-engineered Python program (the data-generation oracle); “Model” is the accuracy the fine-tuned rank- ≤ 32 adapter actually reproduces in greedy CoT, measured on a held-out slice of real training rows (the faithful test oracle). The gap is near-zero for the lookup/arithmetic tasks and enormous for cryptarithm, despite a 0.71 search-solver.

Category	Test wt.	Base	Solver	Model
numeral	16.6%	0.959	1.00	≈ 1.00
unit_conversion	16.8%	0.745	1.00	≈ 1.00
gravity	16.8%	0.585	1.00	≈ 1.00
cipher	16.6%	0.306	1.00	≈ 0.99
bit_manipulation	16.9%	0.080	0.989	0.678
equation_num. (deduce)	6.3%	0.287	0.951	0.83
equation_num. (guess)	1.4%	0.037	0.206 [†]	0.18
cryptarithm (deduce)	6.9%	0.003	0.71 [‡]	0.01–0.07
cryptarithm (guess)	1.7%	0.000	0.59 [‡]	0.02
<i>Overall (best banked adapter)</i>		0.466	,	0.85

[†] The operation is information-free given the symbol, so 0.206 is the marginal-prior ceiling, not a coverage limit. [‡] Search-based solver (backtracking / generate-and-verify). The *forward-derivable* fraction, what a left-to-right CoT could honestly imitate, is ≈ 0 (Section 5.2). Model 95% Wilson CIs (n): bit [.64, .72] (500); eq-deduce wide ($n=18$, see Appendix E); eq-guess [.12, .28] (96); crypt-deduce [.01, .09] (93); crypt-guess [.01, .08] (87); easy-four/cipher $n \approx 1.6k$ (CI $\leq \pm 0.4pp$).

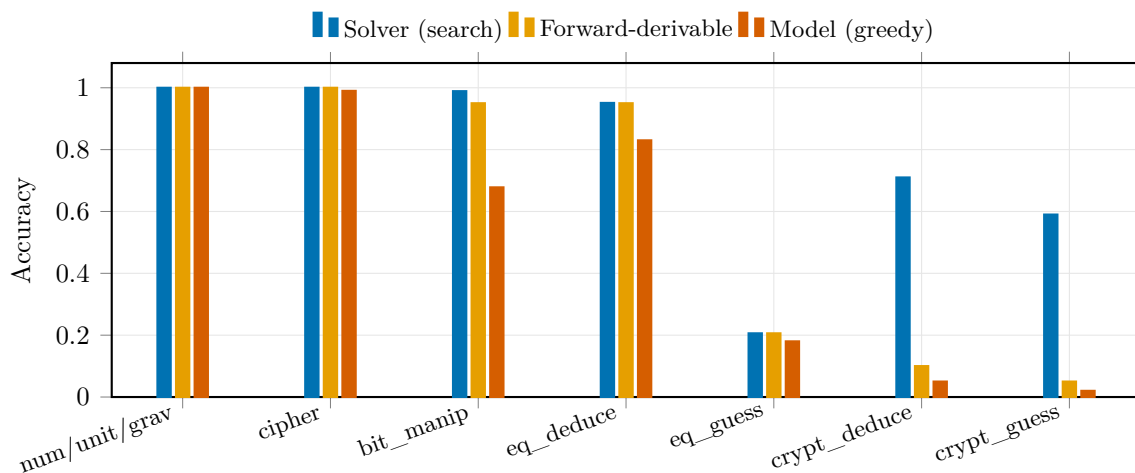


Figure 1: **The learnability frontier.** Solver (backtracking search) vs. the *forward-derivable* ceiling (what a left-to-right CoT could honestly imitate) vs. the fine-tuned model. For lookup/fit tasks all three coincide. For cryptarithm the solver reaches 0.71, but the forward-derivable ceiling collapses to ≈ 0.10 and the model tracks *that*, not the solver (0.05). bit_manipulation is the exception where forward derivation is high yet the model still gaps, a search-depth limit STaR partly closes (Section 5.1). The “guess” tasks are information-limited. Exact values and CIs in Table 1.

by instrumenting the solver to narrate its own steps in the base model’s native ASCII style, ending in a `\boxed{}`; (3) SFT the LoRA on this synthetic corpus. And (4) evaluate the adapter on held-out *real* training rows. Because step (2) uses different values than any evaluation row, there is no leakage; because of the $\text{train} \equiv \text{test}$ property, step (4) is the leaderboard.

I deliberately enforce one CoT discipline throughout: the trace must *show the search* (try $\rightarrow$ reject $\rightarrow$ match) and must never teleport to an answer it has not derived on the page. A trace that states the conclusion without the work teaches nothing transferable; this principle, and its violations, turn out to be the crux of the whole study (Section 7).

Rendering, and what “witnessed” means. I render a CoT by instrumenting the solver to narrate each step in the base’s native ASCII voice: transcribe the relevant glyphs, compute, and emit a *verdict* (“keep,” “drop,” “this run extends”). The one non-negotiable rule is that every verdict must *restate, on its own line, the evidence that forces it, a witnessed verdict*. A verdict that is a fixed phrase decoupled from the line’s own numbers is a *teleport*: it reads as reasoning but carries none. Figure 2 contrasts the two on a real ones-digit elimination; the failure shown is verbatim from a trained model and is the modal cryptarithm error (Figure 5).

Witnessed verdict (rendered) vs. verdict-as-token (emitted)	
WITNESSED:	EQ1 ones: target ones = 4 (= b ones). try a ones 6: 6*4 ends 4 -> MATCH -> keep 6.
TELEPORT:	EQ1 ones: ... 3*4 ends 2; 5*4 ends 0; 6*4 ends 4; 7*4 ends 8; 8*4 ends 2; 9*4 ends 6 -> none matches -> drop.

Figure 2: **Witnessed vs. teleported verdict.** The teleport line is verbatim from a trained model’s transcript (10b71e8a): it computes “6*4 ends 4”, which *matches* the target ones digit 4, on the very same line, then concludes “none matches $\rightarrow$ drop,” wrongly eliminating the correct operation. The arithmetic is correct; the verdict does not follow from it. SFT on traces whose verdicts are not witnessed installs exactly this behavior.

When SFT plateaus on a category I escalate along two axes: RLVR/GRPO against the binary `verify()` reward, and STaR, harvesting the model’s *own* correct rollouts (filtered by the verifier) and folding them back as SFT data. The escalations cost more, so I gate them: a round is only promoted if a cheap held-out probe clears a pre-registered threshold.

Statistical reporting. Accuracies are point estimates reported with Wilson 95% confidence intervals and the evaluation size n . Held-out sets are real-train slices (Section 3): 100/category unless noted, with bit-manipulation evaluated on a disjoint 500-row set and cryptarithm on the $n=93/87$ “unseen” deduce/guess slices. I flag explicitly where small n makes a difference non-significant (it does for the $\text{pass}@k$ comparisons. It does *not* for the headline solver–model gaps).

5 Results

My version-by-version leaderboard trajectory (Appendix F) situates the project. The frozen base model scores 0.466; a null (zero) adapter reproduces it (0.50). Installing the four easy tasks and a first bit-manipulation CoT reaches 0.59. A from-base adapter with finalized cipher/bit/equation grammars banks 0.85. On the final 4,355-team leaderboard 0.85 is the *median*: 2,236 teams reached

≥ 0.85 , only 66 cleared 0.87, and just the leader reached 0.92 (Section 8). My 0.85 is the solver ceiling of the tasks I could make learnable; the missing 0.07 to the top is almost entirely cryptarithm and the bit-manipulation tail. The rest of this section dissects those two.

5.1 Where the procedure transfers: the easy four and bit_manipulation

The four lookup/fit tasks are uninteresting in the best way: once the CoT is correct and ASCII-clean, the model reproduces the solver to ≥ 0.99 . They establish that the pipeline, the budget, and the grader are not the bottleneck.

Bit-manipulation is the informative middle of the frontier, and the place where knowing the generator pays off most. The task’s own prompt names its operations: “*The transformation involves operations like bit shifts, rotations, XOR, AND, OR, NOT, and possibly majority or choice functions.*” This is a gift, it says the hidden rule is a single boolean function over up to three *taps* (rotated/shifted copies of the 8-bit input), drawn from a small human-nameable vocabulary and applied identically to all eight output bits. I make that explicit: a six-function basis {XOR, MAJ, XOR_OR, OR, OR_XNOR, CHOICE} (two 2-argument and four 3-argument functions; Table 2) accounts, under a first-match partition, for 1577 of the 1602 training rows. The remaining 25 need a deeper composition. AND-of-three is redundant, its rows are subsumed by majority (108/109). This is the same single-rule space my global solver searches by op-composition to 98.9% query accuracy (Appendix B). The basis is its interpretable face, mapping one-to-one onto the generator’s stated vocabulary.

The point is the contrast. The *solver* side of bit-manipulation is essentially closed: a handful of named functions over a handful of taps explains the entire dataset. Against that, the fine-tuned model’s 0.678 is not a data problem, it is a learnability problem, and a sharply structured one (Table 4): the rules the model misses are overwhelmingly the three-tap ones (34% of the category, accuracy 0.50), exactly the rules whose search is deepest. The procedure, enumerate the basis $\times$ tap-assignments, verify on all examples, apply, is the bounded search I expected to be teachable. It is, but only in one specific way.

Table 2: A six-function basis for the bit-manipulation rule, mapping onto the generator’s own prompt vocabulary. Each function takes up to three *taps* (independent rotated/shifted copies of the input) and is applied identically to all eight output bits. Counts are a priority-ordered first-match partition of the 1602 training rows.

Function	Arity	Definition over taps (a, b, c)	Rows (first-match)
XOR	2	$a \oplus b$	625
MAJ	3	$(a \wedge b) \vee (a \wedge c) \vee (b \wedge c)$	459
XOR_OR	3	$(a \oplus b) \vee c$	234
OR	2	$a \vee b$	148
OR_XNOR	3	$a \vee \overline{(b \oplus c)}$	58
CHOICE	3	$(a \wedge b) \vee (\bar{a} \wedge c)$	53
<i>covered by basis</i>			1577 / 1602
AND3	3	$a \wedge b \wedge c$	<i>redundant</i> ($\subset$ MAJ, 108/109)

The split is a first-match partition under a fixed priority order; an independent re-derivation confirms XOR (≈ 625 –638) and MAJ (≈ 459 –478) as the two largest and robust strata, while the exact division within the {OR, XOR_OR, OR_XNOR, CHOICE} cluster depends on the ordering (its total does not). The op-composition solver attains 98.9% query accuracy on the same rows (Appendix B).

Every *hand-written* CoT I authored collapsed to ≈ 0.05 –0.06: a per-bit “dialect” (0.046), a hand

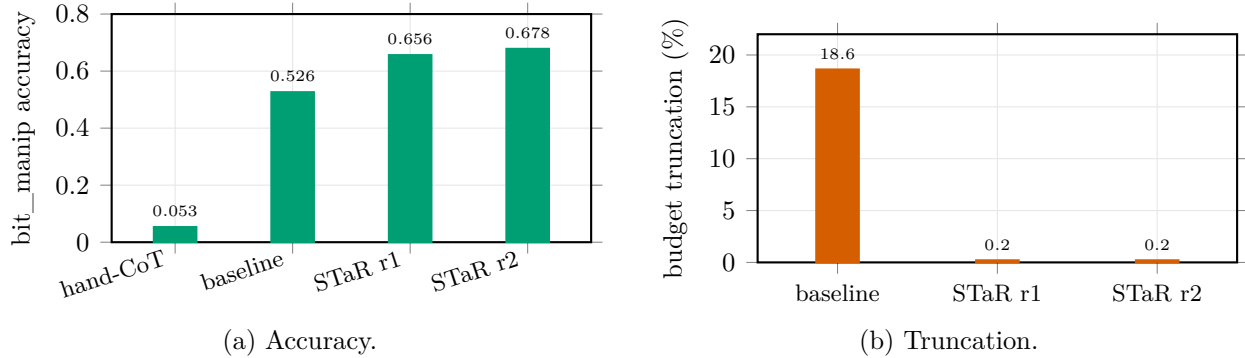


Figure 3: **Bit-manipulation: only the model’s own search transfers.** Hand-written CoT that teleports the rule search scores like the base model (0.053). STaR on verifier-passed self-traces lifts accuracy and collapses budget truncation, because the imitated traces are searches the model can actually run within budget. Baseline→STaR is significant (disjoint 95% CIs, $n=500$). The r1→r2 step is within noise.

grammar that named the taps then asserted the rule (0.062), and a peel-to-two-tap scheme (0.053). The common defect, confirmed by a line-level autopsy of 28 wrong output bits (Section 7), is that each authored trace *teleports* the one step that matters, the rule search, because a human author already knows the answer and cannot un-know it while writing. Transcription, arithmetic, and assembly in these traces are flawless. Only the search verdict is fake.

What worked was STaR. Seeding from a weak warm start, I harvested the model’s *own* verifier-passed rollouts and fine-tuned on them. Accuracy rose $0.526 \rightarrow 0.656 \rightarrow 0.678$ over two rounds, and, tellingly, budget truncation fell from 18.6% to 0.2% (Figure 3). The model’s own successful searches are short, decisive, and (because they actually ran) honest. Imitating them installs a procedure the model can execute at inference, closing the exposure-bias gap that the hand-written traces opened. Accuracy then plateaus: the residual ≈ 0.50 on genuine three-tap rules is the base’s search ceiling on this architecture, and additional synthetic three-tap data did not beat the STaR number ($0.602 < 0.678$).

5.2 Where it does not: cryptarithm

Cryptarithm is the negative result, and it is robust. I ran eleven successive CoT designs (r1–r11; Figure 4 plots the nine that reached a held-out greedy eval, r4 having been gate-failed before training): decode-first verification, truthful conditionals, policy-driven backtracking renderers, two-regime structure modeling, construction-of-symbol-table scaffolds, monotone try-counters with bail-out endings, self-routing forced chains, witness-bearing elimination lines, a generate-and-verify reframe, and explicit terminate-on-exhaustion traces. Every round lands in $[0.01, 0.07]$. The truncation rate swings wildly across rounds, from 87% down to 1%, while accuracy does not move, which already rules out “ran out of budget” as the explanation.

Five walls converge. I then measured the ceiling five independent ways (Table 3). (1) An unbounded full-enumeration vote, the solver with no token limit, answers $457/659 = 0.693$; 25.8% of instances are genuinely ambiguous given the examples. (2) The hidden map is information-free: $MI(\text{digit}; \text{glyph}) = 0.021$ nats versus a shuffle baseline of 0.022, and a battery of structural predictors (ascii-offset, pool-linear, first-appearance order, frequency argmax) all score *at or below* the analytic chance rate of 0.069 on leave-one-digit-out prediction; the two “gap” strata that would require

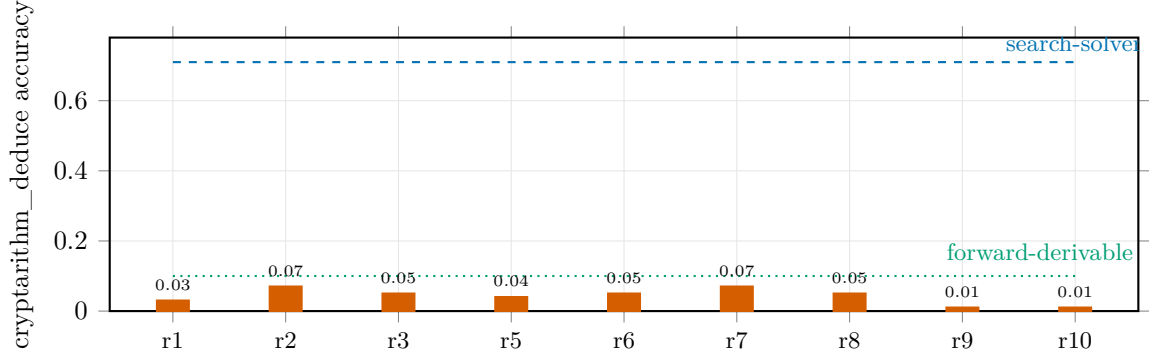


Figure 4: **One floor, many rounds.** Cryptarithm-deduce accuracy under successive CoT designs (r1–r11. The nine reaching a held-out greedy eval are shown; RLVR/GRPO and the generate-and-verify reframe ran between rounds; Table 12 lists all fourteen training runs). The search-solver answers $\approx 71\%$ of instances. The fraction a faithful forward CoT could imitate is $\lesssim 10\%$; the model lands at ≈ 0.05 regardless of design effort.

Table 3: Five independent bounds on cryptarithm-deduce, each a different measured quantity. They converge: the task is not learnable as a forward CoT on this base.

#	Wall (what it measures)	Number
1	Solver ceiling (unbounded enumeration vote)	457/659 = 0.693; 25.8% ambiguous
2	Information floor (map mutual information)	0.021 nats $\approx$ shuffle; gap-strata recover 0
3	Base behavior (native greedy)	0/50 correct; 49/50 truncate
4	Forward derivation (provable-step closure)	1/659 covered
5	Verdict fidelity (per-line audit)	arithmetic 97–100%, verdicts 16–57%

exploiting map structure recover 0 and 0 rows. (3) The base model is structurally lost: native greedy decoding answers 0/50 and truncates on 49/50. (4) The forward derivation is enumeration-locked: a closure that applies only the steps the base provably executes (column arithmetic, injectivity) covers 1/659 instances, and gold maps stall at 0–2 of ~ 10 digits on 449/450 rows. (5) Per-line execution fidelity (next paragraph) caps faithful imitation independently. Crucially these bound *different* quantities: wall 1 bounds the *solver* (any method, ≈ 0.69), walls 1–2 explain why no map-structure shortcut exists, wall 3 bounds the *base*’s native behavior, and walls 4–5 bound the only quantity SFT can actually imitate, the *faithful forward CoT*, near zero. It is that last, binding bound (forward-derivability 1/659 together with verdict fidelity) that pins my runs at ≈ 0.05 ; the solver ceiling merely says a non-forward, search-based method could in principle reach ≈ 0.69 . The convergence is across *kinds* of bound, not one re-measured number.

RLVR is alive but does not transfer. Reinforcement learning against the verifier behaves exactly as the walls predict. On an *eased* curriculum it works: a concat-only tier (pure transcription, no arithmetic) reaches mean reward 0.742 and pass@8 = 15/15, and a smoke test confirms non-degenerate advantages, the gradient mechanism is intact. But on real value-operator instances the reward is 0.000–0.004, and pass@16 on real deduce rows stays near zero across checkpoints (0.16, 0.10, 0.05; $n=20$, 95% CIs ± 11 –15pp and mutually overlapping, no significant trend, only a persistent floor). A budget-unbounded clean pass@32 is 0.125 ($n=40$, CI [0.06, 0.26]). There are essentially no positive rollouts to learn from, so the held-out accuracy never moves: the gradient is

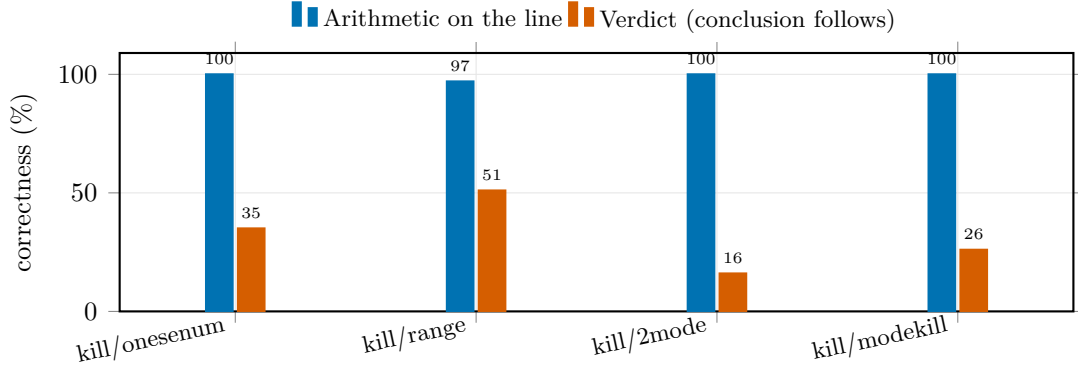


Figure 5: **Verdict-as-token.** Across the elimination-line types that drive cryptarithm, the arithmetic on each line is essentially perfect while the line’s *verdict*, “therefore drop this candidate”, follows from its own numbers only 16–51% of the time. The model imitates the shape of a verifiable step without its content. Line counts $n = 390/102/79/35$. The arithmetic-vs-verdict gap is significant for every type (95% CIs ± 5 –14pp).

present, but it has nothing to grip.

Derivation-blocked, not ranking-blocked. The cleanest single diagnostic is a generate-and-verify reframe: instead of deriving the map, the model proposes hypotheses and a learned prior ranks them. Here $\text{hit@8} = 0.71$ (the gold answer is in the model’s top-8 on 71% of instances) and the worst-case token cost of checking eight hypotheses is well under budget, so *ranking is not the constraint*. When I instead require the model to *derive* each hypothesis by forward arithmetic closure, coverage on the top-8 collapses to 1/659 and the projected yield (0.205) falls below my render gate (0.45), so the round was never trained. The model can recognize a correct map but cannot *construct* one left-to-right, because constructing it requires tracking which assignments are still consistent, explicit search state, across the whole problem.

Verdict-as-token. The mechanism becomes concrete in a per-line fidelity audit of 100 held-out transcripts (7,566 line records), scored on three axes: transcription, arithmetic, and *verdict* (does a line’s stated conclusion follow from its own numbers?). Arithmetic is essentially perfect, the dominant elimination line type (“enumerate ones-residues, none matches, drop this candidate”; $n = 390$) has 100% arithmetic correctness, yet its *verdict* is correct only 34.9% of the time: the model drops candidates that, by its own numbers on the same line, actually match. Other elimination types show the same split ($A=100\%$, $V=16.5\%$; $A=97\%$, $V=51\%$). The median trace derails at its 552nd token. The model has learned the *form* of a verifiable elimination step as a fixed template, decoupled from the logic the step is supposed to encode. Under teacher forcing this template is always emitted in a context where it happened to be correct. At free-running inference it fires unconditionally, a textbook exposure-bias failure [2, 24], but localized to the single most load-bearing token in the procedure.

5.3 The solvability–learnability gap, quantified

Table 1 and Figure 1 make the headline claim precise. The correlation between solver accuracy and model accuracy is high among the lookup/fit tasks and *breaks* exactly where the procedure stops being a forward computation and becomes a search: bit-manipulation (solver 0.99, model 0.68) and

cryptarithm-deduce (search-solver 0.71, model 0.05). The information-limited “guess” tasks are the control: there the solver itself is low (the answer is not determined by the prompt), and the model tracks it closely (0.18 vs 0.21; 0.02 vs 0.59 where the guess is bottlenecked by the upstream cryptarithm search). Solver coverage, in short, predicts model accuracy only when the solver is a forward map. For search procedures it predicts almost nothing, unless the search is removed from the trace by memorizing its structure (Section 9).

6 Anatomy of the Failures

I characterize the failures at the line and token level across all nine tasks (Table 5. Full per-sub-category numbers in Appendix E). Two facts frame everything. First, *for the shipped adapters, failure is wrong-answer, not budget*: cryptarithm-deduce finishes within budget and is wrong on $\approx 96\%$ of instances (1% truncated), and bit-manipulation truncates on 0% (17% finished-but-wrong). The levers that matter are correctness levers, not length levers. The one exception is diagnostic in itself: the rendered cryptarithm *search* traces (backtracking narrated step by step) loop and truncate on 64–72% of instances, because the procedure needs a tried-set the model does not maintain. Second, *the errors are not arithmetic*: a 28-output-bit autopsy and a 7,566-line cryptarithm fidelity audit both find transcription and arithmetic essentially correct, what fails is the *decision*: which rule, which verdict.

Bit-manipulation: the failures are rule-selection on hard taps. Accuracy is a clean function of tap count (Table 4): one- and two-tap rules are largely solved, three-tap rules sit at chance-corrected 0.50. An autopsy of 28 wrong output bits across 12 transcripts found *zero* transcription or arithmetic errors; 21 were fixable rule-selection slips (a dropped shift-anchor that shifts a run by one, run accept/reject steps whose lines did not restate the evidence that should have decided them, and tie-breaks), and 7 were genuine three-tap bits for which no ≤ 2 -input rule exists. Bit accuracy is therefore bounded by ≈ 0.82 in any per-bit grammar and by ≈ 0.50 on three-tap rows in any grammar the base can actually execute, which is exactly where STaR plateaus.

Table 4: Bit-manipulation accuracy by tap count (best adapter, disjoint 500-row eval). Difficulty is monotone in the number of input taps the rule combines; the three-tap stratum is the residual ceiling.

Rule complexity	n	Accuracy (95% CI)	Share
1 tap	42	0.881 [.75, .95]	small
2 taps	283	0.763 [.71, .81]	$\approx 56\%$
3 taps	169	0.503 [.43, .58]	$\approx 34\%$
overall	500	0.678 [.64, .72]	,

Cryptarithm: the verdict is decoupled from the evidence. The fidelity audit (Figure 5) localizes the failure to elimination *verdicts*. The signature is concrete and reproducible: the model enumerates candidate residue pairs, *writes one that matches the target on the same line*, and then concludes “none matches $\rightarrow$ drop.” The first false line arrives at a median of token 552. When I instead render the search explicitly so the model must run it, it either loops forever (69% of the generate-and-verify round re-enter an identical attempt, having no record of what was tried) or,

when forced to terminate, teleports to a guess (31%, correct 1/31). Both are the absence of carried search state, seen from two sides.

The native-voice attractor (a starkest-case exposure bias). A distinct and, I think, broadly important mode surfaced when I authored grammars in a *non-native* voice (traces opening with `Bit.../Group...` rather than the base’s habitual `I...`). Teacher-forced fidelity on these grammars was 0.92–0.95, the model demonstrably *learned* them, yet at greedy decoding it entered them 0% *of the time*: at the very first generated token it emits its native opener instead, a 15–16 nat divergence, and never reaches the learned procedure. The grammar is a fully-learned but unreachable island. Cryptarithm, whose traces happen to share the native opener, does enter the grammar and then *drifts* mid-trace, which is how it manages to score *below* a constant-output baseline (destructive interference with the base’s own attempt). The operational lesson is sharp: author chain-of-thought in the base’s native voice, or it will never be decoded at inference no matter how well it is fit. This is exposure bias [2, 24] at its most extreme, the training and free-running distributions diverge at token 0.

Table 5: The nine observed failure modes collapse to four root causes; the first, a verdict decoupled from its own evidence, dominates, and is the unfaithful-CoT signature [17, 28] in the training regime.

Root cause	Variants (dominant category)	Measured signature
1. Verdict decoupled from evidence (“verdict-as-token”)	teleport / unwitnessed claim. Verdict-as-constant; fabrication-on-exhaustion; constant-concat rubber-stamp (crypt; authored bit)	verdict fidelity 16–51% vs arithmetic 97–100%; SFT rounds capped 0.03–0.07
2. No carried search state	infinite loop / no tried-set. Budget-truncating search renders (crypt)	re-enters identical attempts; loops to 7680-tok cap on 64–72%
3. Distribution mismatch (exposure bias)	never-enters / anchoring. Mid-trace drift (non-native grammars; crypt)	teacher-forced fidelity 0.95 yet greedy adoption 0.0; diverges at/after token 0
4. Output corruption	symbol-table hallucination (crypt encode)	correct derivation, glyphs emitted out of order (T fidelity 51%)

7 Analysis: Why Some Procedures Resist Distillation

My results compose into one account, in three threads. *Thread 1* is a controlled refutation: distilling the search fails on every backbone, not just the hybrid one (every backbone caps). *Thread 2* identifies the failure mechanism: SFT installs the surface form of a verifiable step, a *verdict decoupled from its evidence*. *Thread 3* gives the root reason: the search admits no faithful forward chain-of-thought to imitate. The chain is: no faithful forward trace $\Rightarrow$ SFT copies only the shape $\Rightarrow$ verdict-as-token $\Rightarrow$ collapse at free-running inference, on every backbone. The dual escape, memorizing the search’s finite structure rather than distilling the search, is taken up in Section 9.

1. The bottleneck is the task, not the architecture. The base model has every primitive cryptarithm needs, it does column arithmetic correctly 97–100% of the time, recognizes a correct map 71% of the time when one is proposed, and handles the purely transcriptional concat family. What it cannot do is run a search: maintain the set of tried assignments, propagate constraints,

and decide a branch is dead. I first suspected the hybrid Mamba-2/MoE backbone: state-space layers compress history into a fixed-size state [4] that sits in TC^0 and cannot perform genuine sequential state-tracking [20], a natural reason a growing search frontier cannot be carried. A controlled architecture sweep *refutes* that explanation for this task (Table 6): trained on the identical cryptarithm corpus, two dense Transformers (Llama-3.2-3B, Qwen3.5-4B) and an MoE Transformer (gpt-oss-20b) hit the *same* ≤ 0.04 floor, each fits the corpus (train NLL $\rightarrow$ floor) yet reproduces almost none of it at inference. The search-distillation failure is therefore *architecture-general*, and the binding constraint is task-intrinsic, no faithful forward CoT to imitate (thread 3), not the SSM backbone. (State compression may still contribute to the bit-manipulation three-tap tail, where a rank- ≤ 32 adapter is too small a perturbation to install a search memory.) The same holds *without* fine-tuning: sampled in-context under the competition’s 7680-token budget, DeepSeek-V3.1 (671B) scores 0.05 (finishes, wrong) and Nemotron-Super-120B scores 0.00 (truncates on 100%), so it spans 3B $\rightarrow$ 671B and SFT $\rightarrow$ in-context alike (Table 6). The bit autopsy is the same story in miniature: of 28 wrong output bits, 0 were transcription or arithmetic errors. All traced to run-extension *verdicts* (accept/reject/tie-break) whose lines did not restate the evidence that should have decided them.

Table 6: **The search does not distill, on any backbone.** Cryptarithm-deduce accuracy, all under the competition’s 7680-token budget. *Top*: the identical corpus fine-tuned (rank-32 LoRA, 4 epochs) on four backbones, 3B–30B. *Bottom*: two frontier models sampled *in-context* (no fine-tuning. The prompt already carries the worked examples). Spanning 3B $\rightarrow$ 671B and fine-tuned $\rightarrow$ in-context, none reproduces the search. At frontier scale the failure splits: V3.1 finishes but is wrong. The 120B truncates on 100% within budget. The exact 30B/A3B Transformer match (Qwen3-30B-A3B) was unschedulable on my compute.

Model	Params (tot/act)	Architecture	deduce	note
<i>Fine-tuned (rank-32 LoRA, identical corpus; n=100)</i>				
Nemotron-3-Nano	30B / 3.5B	Hybrid Mamba-2+MoE	0.03–0.05	
Llama-3.2-3B	3B / 3B	Dense Transformer	0.010	
Qwen3.5-4B	4B / 4B	Dense Transformer	0.040	
gpt-oss-20b	21B / 3.6B	MoE Transformer	0.010	
<i>In-context, no fine-tuning (raw base; n=20)</i>				
DeepSeek-V3.1	671B / 37B	MoE Transformer	0.050	finishes, wrong
Nemotron-Super	120B / 12B	Hybrid Mamba+MoE	0.000	100% truncate

2. Verdict-as-token is the failure’s signature under teacher forcing. A verifiable solver’s most important lines are its verdicts, “this candidate is impossible,” “this run extends.” When a CoT is written so that a verdict is a fixed phrase rather than a function of locally-written evidence, SFT learns the phrase, not the function, because under teacher forcing the phrase is always seen in a correct context. The defect is invisible in training loss, across fourteen cryptarithm training rounds the train NLL converged to the floor (e.g. crypt-r1 reaches NLL 0.146 by step 35 and ≈ 0.001 by the end; the Kaggle merge runs log a final loss ≈ 0.003) while held-out deduce accuracy never left [0.01, 0.07] (Table 12), and invisible under teacher-forced evaluation. It appears only at free-running greedy decode, as the verdict firing in the wrong context. This is why every *authored* trace failed and only *self-harvested* traces (STaR) succeeded for bit-manipulation: a self-trace’s verdict is, by construction, one the model produced *and the verifier confirmed*, so the evidence $\rightarrow$ verdict link is real rather than narrated.

3. The root cause: no faithful forward CoT to imitate. Cryptarithm is the extreme case: the hidden map is information-free (wall 2), so there is no left-to-right derivation that a human or solver could write down that arrives at the map by forward reasoning, the only honest route is backtracking search, which forward CoT cannot represent without carried state. The forward-closure coverage of 1/659 is the formal statement: the set of instances solvable by the steps the base can actually execute, in order, is empty for all practical purposes. When no faithful forward CoT exists, SFT can only learn an unfaithful one, and the unfaithful one is exactly the verdict-as-token template that collapses at inference.

A predictive rule of thumb. Together these suggest a cheap pre-test for whether a solvable task will be CoT-learnable on a small model, applied *before* any training: (a) does a purely forward closure (no backtracking, no global constraint propagation) cover a meaningful fraction of instances? (b) does the task’s hidden structure carry non-trivial mutual information with observable tokens? (c) does the base model, prompted natively, attempt the right *kind* of move rather than truncating? Cryptarithm fails all three; bit-manipulation passes (a) and (c) and is learnable via the model’s own search; the easy four pass trivially. I derived this *post-hoc* from the nine tasks here, but test its core claim, that forward-derivability is the causal lever, prospectively next (Table 7); the broader three-part heuristic remains a hypothesis I offer for testing.

A prospective test: the forward-derivability intervention. To check the core claim causally rather than by correlation, I took the *same* cryptarithm instances and revealed a fraction of the digit↔symbol key in the prompt, turning that fraction of the derivation from search into forward lookup, then fine-tuned Nemotron on each (Table 7; $n=42$). With no key (the original task) accuracy is 0.03; revealing *half* the key barely moves it (0.048). Revealing the *full* key lifts it to **0.571**, a $\sim 15\times$ jump on an otherwise identical task (disjoint 95% CIs), with the residual gap to 1.0 explained by ordinary op-execution errors (cf. equation_numeric at 0.83). Two things follow. First, forward-derivability is *causal*: it is the lever, holding the surface task fixed. Second, the effect is sharply *nonlinear*, half-forward does not give half-accuracy, because any residual search re-introduces the verdict-as-token failure; a faithful forward CoT must cover the *whole* derivation, not most of it.

Table 7: **Forward-derivability is causal.** The *same* cryptarithm instances, with a fraction of the cipher key revealed in the prompt (turning that fraction of the derivation forward), fine-tuned on Nemotron ($n=42$, Wilson 95% CI). Making the task fully forward lifts accuracy $\sim 15\times$. Half-forward barely helps, any residual search re-triggers the collapse.

Key revealed	Forward-derivable	crypt-deduce
none (original task)	≈ 0	0.03
half	$\approx 50\%$	0.048 [.01, .16]
full	100%	0.571 [.42, .71]

8 Competition Dynamics and External Corroboration

8.1 Leaderboard structure and external corroboration

The benchmark is drawn from a public competition [14], whose final 4,355-team leaderboard is itself evidence for my thesis. The score distribution has a sharp plateau exactly where I place

the frontier: 2,236 teams reached ≥ 0.85 , only 66 cleared 0.87, 7 cleared 0.88, and just two teams reached ≥ 0.90 , with a single leader at 0.92. A score of 0.85 is the *median*. This is the shape my analysis predicts: the easy four plus a learnable share of bit-manipulation get a competent team to ≈ 0.85 , and the last few points require cracking cryptarithm, which almost no one did.

Independent corroboration of the ceiling. The competition’s Open Progress Prize (awarded for the best reproducible *methodology*, separately from the raw leaderboard) went to a fully open-source solution [26, 27] that, independently of me, took the *same* approach, reverse-engineer each generator, then SFT a rank-32 LoRA on solver-grounded synthetic chain-of-thought (LoRA rank 32, adapt MLP/attention/unembed, LR 2×10^{-4} , single epoch, 8192 context, category-stratified batching). It also scored **0.85**. Two independent teams converging on the same method *and* the same ceiling is strong evidence that 0.85 is the ceiling of the *search-distillation approach* both used, not of either team’s execution; the 1st-place solution’s memorization–computation reformulation breaks past it to 0.92 (Section 9). My training recipe and theirs match closely (Appendix C), which is no coincidence: ours was tuned toward the same proven regime.

The residual is generator hardness, not modeling laziness. My own frontier baseline agrees: DeepSeek-V3.1 (671B) and Nemotron-Super-120B, prompted in-context on cryptarithm, score 0.05 and 0.00 (Table 6). And a third-party effort reports several hundred of the hardest puzzles *unsolved even by frontier teacher models* given the recovered rule at high reasoning effort. The ceiling is in distilling the search, not in a failure to try.

8.2 Threats to validity

Three are worth recording. (1) *The metric has a binary-string trap*: binary answers are compared as exact strings, so a correct value at the wrong width ("11011" vs "00011011") is scored wrong; this is a hazard to avoid (emit the exact width), not a lever. (2) *Public/private overfit*: competitors discussed choosing a cryptarithm default operation (reverse- vs. forward-concatenation) that raised the *public* score while *lowering* the private one, a leaderboard-specific artifact, and a caution for anyone reading single-split numbers. (3) *Memorization under train $\equiv$ test*: with train and test from one generator, multi-epoch training can drive the training metric to ceiling by memorizing rather than learning the procedure, which is precisely why I train on synthetic instances with *different* values and report held-out (“unseen”) accuracy rather than the contaminated headline (Appendix E). A concrete leaderboard-gaming technique I can *document* rather than merely report is **adapter-noise laundering**: a notebook in my own repository loads a *third party’s* shared submission adapter and adds Gaussian noise at 20% of each tensor’s standard deviation to all 12,010 weight tensors, then repackages it as a fresh submission. The perturbation is small enough to preserve most of the donor’s accuracy while yielding a nominally distinct artifact. I did not measure its leaderboard effect (my copy records no score), so I report the *procedure* as a reproducible integrity concern, not a result, a shared high-scoring adapter can be re-skinned into many ostensibly independent submissions.

My own public/private near-miss (verified). The selection risk was not hypothetical for me. My full submission record (Kaggle API; Table 8) shows my best *private* score was **0.860**, a native-HuggingFace run (EXP-138, forking Tong’s published recipe [27]) trained on the 0.86-adapter’s verifier-passed traces plus 1,600 synthetic bit puzzles, yet that submission scored only 0.844 *public*, while my public-best submissions sat at 0.852–0.856. The trap is sharpest in the reference adapter I benchmarked: 0.856 public but 0.832 private (it overfit the public split). Standard public-LB

selection would have discarded my 0.860-private model and kept a weaker one. The public score, computed on a few-hundred-row split, is a lossy selector. A genuinely stronger model can go unselected, which is how a medal-class private result was left on the table.

Table 8: Public vs. private leaderboard for my key submissions (Kaggle API record). The ordering inverts: my best *private* score (0.860, EXP-138) is mid-pack on *public*, while the best *public* score (the reference adapter, 0.856) is among the worst on private. Selecting on the public column picks the wrong submission.

Submission	public	private
EXP-138 (native-HF, 06-05)	0.844	0.860
v18 (+ bit STaR)	0.848	0.856
(06-06)	0.840	0.852
v17 (banked)	0.852	0.852
v15	0.828	0.840
reference adapter (≈ 0.86)	0.856	0.832
v6_7	0.768	0.768

9 Memorization, not search

The search-distillation ceiling documented above is a ceiling on *distilling the search*, not on the task. The competition’s 1st-place solution (team NullSira; Private LB 0.920) reaches high cryptarithm and bit-manipulation accuracy by an approach its authors describe as deciding “what the model should memorize ... and what it should compute inside the trace” [8], and it is the cleanest illustration of the present thesis: it succeeds precisely by *not* teaching the model to search.

The naive search over a cryptarithm prompt is intractable to narrate, up to $10!$ digit assignments times 24^3 operator rules ($\approx 5 \times 10^{10}$ candidates), which cannot be enumerated within the 7680-token budget. The winning solution removes that search from the trace. For a single equation it precomputes a *signature*, the pattern of repeated symbols normalized by first occurrence (e.g. ABCCDD), and enumerates all two-digit $\times$ two-digit operands under the 22 non-join rules into a catalog of 4,205 signatures, each mapping to a short list of candidate (rule, digit-assignment) rows with counts. The model *memorizes this catalog* through heavy SFT; at inference it recalls the candidate list for each equation rather than deriving it. The trace then does only the cheap residual: pick the equation with the fewest candidates as an anchor, and run a bounded DFS that checks the recalled candidates against the remaining equations (same symbol $\rightarrow$ same digit, same operator $\rightarrow$ same rule), pruning by mode and group constraints. The combinatorially hard step, candidate generation for the first equation, is memorized; only a short, bounded verification remains as chain-of-thought.

Bit-manipulation is handled the same way: 5,238 valid 8-rule sequences (every coherent 8-bit transformation, normalized to per-output-bit rules) are precomputed and memorized, the greedy per-bit search is projected onto the nearest catalog sequence under Hamming distance, and candidates are verified against the examples (bit columns HEX-compressed to fit the budget). In both cases the pattern is the dual of my negative result. Distilling the search fails because no faithful forward trace exists (verdict-as-token); the escape is to compute the search *offline* into a finite catalog, store it in the weights as memorized lookup, and leave only recall plus bounded verification in the trace. What transfers is therefore not search but *memorization and verification*. This sharpens the boundary beyond “solvable vs. learnable”: a verifiable search is learnable as a chain-of-thought ex-

actly when its candidate structure compresses to a catalog small enough to memorize and a residual cheap enough to verify in-budget. (The winners’ solver-coverage figures include train-set leakage by their own account, but their 91.57% held-out validation, on the same train $\equiv$ test generators I use, confirms generalization within the distribution rather than row memorization.)

10 Limitations

My study is deep on one benchmark rather than broad. I *did* run a controlled architecture sweep (Table 6): training the identical cryptarithm corpus on dense Transformers (Llama-3.2-3B, Qwen3.5-4B) and an MoE Transformer (gpt-oss-20b) yields the same ≤ 0.04 floor as the hybrid, so the search-distillation ceiling is architecture-general rather than Mamba-specific. On scale, frontier models up to 671B sampled in-context under the competition budget also cap (0.05 / 0.00, Table 6). The one cell I could not run is a frontier model *fine-tuned* at length (and the exact 30B/A3B Transformer match, unschedulable on my compute). I also report single training seeds per cell. The binomial CIs bound sampling noise but not seed variance. I evaluate under greedy decoding because the grader does; sampling-based test-time search would change the achievable numbers (my hit@8 and pass@32 results quantify by how much) but not the single-trace learnability question I study. Some internal figures I deliberately did not promote into the headline claims because they appear in my run ledger but not in a re-runnable report (e.g. a pass@32 value attached to a specific value-rule count); I cite only the report-backed numbers and flag this in my released notes. Finally, I did not win the competition: my best *public* score was 0.852 and my best *private* score 0.860, medal-class, but on a submission public-LB selection would not have chosen (Section 8); the leader reached 0.92. I regard that as orthogonal to the contribution, the value here is the instrumented account of the frontier, not the rank. And the cryptarithm ceiling *does* yield to an idea I did not pursue: the memorization–computation split of the 1st-place solution (Section 9). My contribution is the mechanistic account of why distilling the search fails, which is precisely why that split is necessary.

11 Conclusion

I set out to convert perfect symbolic solvers into a small model’s chain-of-thought, on a benchmark engineered so that held-out training accuracy is the test score. Most tasks transferred. One did not, and its resistance was not a matter of effort: eleven CoT designs, RLVR, and STaR all left cryptarithm at ≈ 0.05 , and five independent measurements agreed on why. The unifying lesson is that a verifiable *search* is not a learnable chain-of-thought. What a small autoregressive model can imitate is a forward computation; what a search procedure requires is carried state, a record of what has been tried and a decision about when to stop, and when a task’s only honest solution is search, SFT can learn only the *shape* of the solver’s verdicts, which then misfire at inference. There are two ways across this gap, and both avoid distilling the search itself: teach the model a search it can actually run (bit-manipulation, via STaR on its own successful traces), or remove the search from the trace entirely, precomputing its finite structure into a catalog the model memorizes and verifies against (Section 9; the 1st-place solution). The right way to read solver coverage, then, is as a necessary but badly insufficient condition: it tells you the answer exists, not that your model can reason its way to it, and least of all that the solver’s *search* will become its chain-of-thought.

Ethics and Broader Impact

This work is diagnostic and defensive: it introduces no new model capability and no attack on a deployed system. It does surface competition-integrity concerns the community should weigh: a public/private leaderboard split makes the public score a lossy selector (Section 8), and a shared high-scoring adapter can be trivially perturbed (“adapter-noise laundering”) into many ostensibly independent submissions. I report these to inform benchmark and competition design (private-split selection, submission provenance), not to enable gaming, I did not submit the noised adapter as my own work. Finally, the deterministic-generator benchmark is a controlled proxy for procedural reasoning; my findings should not be read as a claim about a model’s general capability.

Reproducibility and Artifacts

Code, paper source, the architecture-control scripts, and per-row eval data are released at <https://github.com/harshpatel1692/search-not-learnable> (interactive walkthrough at <https://nemotron.harshpatel.live>). I release, per category: the reverse-engineered solvers and the synthetic-CoT renderers; the held-out evaluation harness; the per-row eval CSVs behind the figures; and the per-line fidelity audit (7,566 records). The benchmark data itself belongs to the competition and is available from Kaggle, not redistributed here. Numbers in this paper are drawn from a dated experiment log. Where a figure exists only in my run ledger and not in a re-runnable report, I say so rather than promote it.

A Solver Pseudocode

Each solver is the data-generation oracle for its category: it recovers the hidden rule from the worked examples and applies it to the query. “Coverage” is the fraction of the 9,500 training rows the solver answers correctly. I give the four lookup/fit solvers compactly, then the two search solvers (equation and cryptarithm) in full; the bit-manipulation solver is in Appendix B.

The four forward solvers (numeral, unit, gravity, cipher). *numeral* (100%) is a fixed greedy subtractive encoder over the 13 Roman value-symbol pairs, it ignores the examples entirely. *unit_conversion* (100%) fits the linear map $y = ax + b$ from the two most-separated example points (for numerical stability under the grader’s 10^{-2} tolerance) and applies it. *gravity* (100%) fits the single constant $A = d/t^2$ from one example of $d = \frac{1}{2}gt^2$ and applies At_*^2 . Only *cipher* requires search, over a *closed* vocabulary; I give it in Algorithm 1.

Equation_numeric (deduce 95.1%, guess 20.6%). Each example is $ABgCD = \text{RHS}$ with *visible* digits. The row has one global string mode (identity or reverse) and each operator glyph g carries one operation from a 13-op library (Algorithm 2). The policy locks the query operator to the first op in a frequency-ordered list consistent with *all* of that glyph’s examples, then accepts the mode only if every other glyph is also explainable under it. For an unseen query glyph (“guess”) the operation is undeducible, so the policy pins the mode from the known glyphs and bets the marginal-prior operation (**sub**), which is exactly the 20.6% marginal ceiling.

Cryptarithm (deduce: policy 67.7%, posterior vote 70.9%). This is the search solver, and the contrast between it and what a forward CoT can imitate is the heart of the paper. The hidden per-row digit→symbol cipher is information-free, so no column reveals a digit. The map is pinned

Algorithm 1 CIPHER, monoalphabetic substitution over a closed 77-word vocabulary (100%)

Require: example pairs $\{(c_i, p_i)\}$ (cipher word $\rightarrow$ plain word); query cipher words Q ; global plaintext vocabulary V , $|V| = 77$

```
1:  $m \leftarrow \{\}$ ;  $inv \leftarrow \{\}$   $\triangleright$  cipher $\rightarrow$ plain letter map and its inverse
2: for each pair  $(c_i, p_i)$  with  $|c_i| = |p_i|$  do  $\triangleright$  seed from equal-length alignments
3:   for each position  $j$  do  $m[c_i[j]] \leftarrow p_i[j]$ ;  $inv[p_i[j]] \leftarrow c_i[j]$ 
4: repeat  $\triangleright$  constraint propagation to fixpoint
5:   for each query word  $w \in Q$  not fully decided do
6:      $cand \leftarrow \{u \in V : |u| = |w|, u \text{ consistent with } m, inv, \text{ and intra-word injective}\}$ 
7:     if  $|\{u : u \in cand\}| = 1$  then lock that word's letters into  $m, inv$ 
8: until no change
9: return  $\bigsqcup_{w \in Q} \text{decode}(w, m)$   $\triangleright$  unresolved letters render as ?
```

Algorithm 2 EQUATION_NUMERIC, policy_seq (deduce 95.1%, guess 20.6%)

Require: examples grouped by glyph g ; query (qa, qg, qb) ; op library $\mathcal{O}$ (13 ops). Frequency order SEQ; default sign-render table

```
1: for  $mode \in (\text{REV}, \text{ID})$  do  $\triangleright$  reverse first: generator prior 476:239
2:    $qop \leftarrow$  first  $o \in \text{SEQ}$  consistent with all examples of  $qg$  under  $mode$ 
3:   if  $qg$  unseen then  $qop \leftarrow \text{sub}$   $\triangleright$  guess: marginal-prior bet
4:   if every other glyph  $g'$  also has some consistent op under  $mode$  then  $\triangleright$  global-mode check
5:     return  $\text{RENDER}(qa \ qop \ qb, mode, \text{SIGNPOS}(mode, qop))$ 
```

only by a backtracking search that requires *all* example equations to hold simultaneously, under the right per-glyph operation (from a ~ 30 -op vocabulary), reading direction, and sign convention (Algorithm 3). Two sound prunes (units-digit congruence and full-operand forcing with injectivity) and a most-constrained variable order keep the $10!$ assignment space tractable; the solver enumerates up to 3000 consistent assignments and scores each under a two-regime (canonical + scrambled) mixture prior fit by EM, then returns the answer with maximum posterior mass. Crucially, the *forward-only* variant of this procedure (constraint propagation with no backtracking, Algorithm 4) is what a left-to-right CoT could honestly reproduce, and it covers $\approx 1/659$ of instances, which is why every supervised attempt to teach the search collapses (Section 5.2).

B Bit-Manipulation Solver, Basis, and Renderer

Global single-rule search (98.88%). The production solver (Algorithm 5) treats the rule as one boolean function over up to three *taps* and searches a composed expression grammar rather than the named basis directly. A *tap* is a transformed copy of the 8-bit input, ROT/SHL/SHR by $k \in \{0, \dots, 7\}$, shifts zero-filling at the boundary, giving 22 taps; NOT is folded into the grammar. The grammar enumerates boolean expressions over three projection variables by op-composition to depth 3 (ten two-input gates plus unary NOT), deduped by 8-bit truth table. For each expression the solver tries every assignment of distinct taps to its variables (a single var: 22; two: $22 \cdot 21$; three: $\text{perm}(22, 3) = 9240$), fail-fast on output bit 0 before checking bits 1–7, and accepts the first assignment that reproduces *all* eight bits of *every* example; it then applies that rule to the query. On the cache of all 1602 rows the solver finds a consistent rule for every row but only 1584 (98.88%) generalize to the query, the 18-row gap is genuine over-fitting (a rule consistent on the

Algorithm 3 CRYPTARITHM, search solver `cryptarithm2` (deduce vote 70.9%)

Require: example equations $\{(L_i, R_i)\}$, each LHS = $s_0 s_1$ OP $s_2 s_3$. Query LHS; op vocab $\mathcal{O}$ (~ 30); EM priors

```
1:  $mass \leftarrow \{\}$ 
2: for  $(rev, radix) \in \{(\perp, 10), (\top, 10), (\perp, 11), (\top, 11)\}$  do  $\triangleright$  radix 11 only as fallback
3:   normalize rows for  $rev$  (reverse magnitudes/result. Detect sign glyph); reject if a sign glyph
   sits inside a magnitude
4:   per glyph, restrict candidate ops by sign pattern and operand magnitude bounds. If all
   rows match a structural concat, fix concat
5:    $\mathcal{S} \leftarrow$  DFS over digit $\rightarrow$ symbol assignments (most-shared symbol first), pruned by units-digit
   congruence and full-operand forcing + injectivity, up to 3000 solutions
6:   for each solution  $\sigma \in \mathcal{S}$  do
7:      $score \leftarrow$  logsumexp of canonical-regime and scrambled-regime log-priors of  $\sigma$ 's op assign-
     ment
8:   render the query answer under  $\sigma$ ;  $mass[answer] += e^{score}$ 
9: return  $\arg \max_a mass[a]$   $\triangleright$  posterior-mass vote. Ties by max score then lexicographic
```

Algorithm 4 CRYPTARITHM-FORWARD, the imitable forward closure ($\approx 1/659$ coverage)

Require: committed (op, rev) per glyph; example equations

```
1: initialize every symbol's digit domain to  $\{0, \dots, 9\}$   $\triangleright$  no teleporting
2: repeat
3:   units mod 10: narrow domains via  $(x \circ y) \bmod 10$  on placed units symbols
4:   magnitude bracket: narrow via result length / value range
5:   all-different cascade: remove singletons from other domains
6:   per-equation GAC: keep only operand/result digits that appear in some consistent  $(a, b)$ 
7: until no change or stalled
8: if every query operand symbol has a singleton domain then return forward-derived answer
9: else return STALLED  $\triangleright$  the common case:  $\approx 658/659$ 
```

shown examples that predicts the wrong query output). The same single-rule space, re-described in human terms, is the basis of Table 2; “coverage by the basis” ($\approx 98\%$ *fit on examples*) and “98.88%” (*correct on the held-out query*) are different measurements and should not be conflated.

Algorithm 5 BIT-MANIPULATION, global single-rule search (98.88%, 1584/1602)

Require: example pairs $\{(in_i, out_i)\}$ of 8-bit strings; query input x_*

- 1: $\mathcal{T} \leftarrow$ the 22 taps $\{(\text{ROT}, 0)\} \cup \{(\text{OP}, k) : \text{OP} \in \{\text{ROT}, \text{SHL}, \text{SHR}\}, k = 1..7\}$
 - 2: $\mathcal{G} \leftarrow$ boolean expressions over vars $\{A, B, C\}$, composed to depth 3 (10 binary gates + NOT), deduped by truth table
 - 3: **for** each expression $e \in \mathcal{G}$ in increasing depth **do**
 - 4: **for** each assignment of distinct taps in $\mathcal{T}$ to the variables *used* by e **do**
 - 5: **if** e reproduces output bit 0 of all examples **then** $\triangleright$ *fail-fast*
 - 6: **if** e reproduces *all* 8 bits of all examples **then**
 - 7: **return** apply e (with this tap assignment) to x_*
 - 8: **return** NONE $\triangleright \approx 1\%$: *under-constrained* / *>3-tap tail*; 5 s time limit
-

Tap-count distribution (from the solve cache). The 1602 recovered rules split 0-tap 5, 1-tap 149, 2-tap 899 (56%), 3-tap 549 (34%). The two- and three-tap shares are exactly the strata in the model’s accuracy breakdown (Table 4). The three-tap stratum is both the largest hard slice and the residual ceiling. The 18 over-fit failures (a rule consistent on the examples but wrong on the query) span 4 one-tap, 7 two-tap, and 7 three-tap rows.

Verifiable rule inventory. Table 9 is the op-composition taxonomy of all 1602 recovered rules, read directly off the solve cache, the machine-checkable complement to the human-named basis of Table 2. The two views agree on the shape (XOR dominant, a large XNOR stratum corresponding to the basis’s OR_XNOR/CHOICE families, identity/copy rules, and AND/OR combinations); the named basis is the interpretable face, this table is the reproducible ground truth.

Table 9: The op-composition rule inventory over all 1602 training rows, computed directly from the solve cache (root operator of the recovered expression; multi-input rules compose these). Taps are near-uniform over rotate/shift by $k \in \{1, \dots, 7\}$.

Root operator	Rows	Tap count	Rows
XOR	444	0 taps	5
OR	350	1 tap	149
XNOR	288	2 taps	899
AND	187	3 taps	549
identity / copy	149		
NOT_A_AND_B	96	tap type	
NOT_A_OR_B	79	SHL	1253
constant	5	SHR	1197
NOR	4	ROT	1144

Per-bit stride renderer (84.6%, used only to author CoT). To *narrate* a search the model could imitate, a second solver expresses each output bit j as a constant, a (possibly negated) copy of one input bit, or a two-input gate of two input bits, with the referenced input positions advancing

by a fixed stride as j increases (because taps are rotations/shifts). It greedily set-covers the eight output bits by maximal cyclic runs, preferring longer runs and simpler descriptors. Output bits needing three inputs are left uncovered and default to 1, the source of its 84.6% ceiling. This renderer is a CoT authoring tool, not the production solver. As Section 5.1 reports, hand-authored narrations of either solver fail to transfer, and only STaR on the model’s own searches does.

C Training Configuration and Submission History

I used two training stacks. The early “lean” lineage (HuggingFace `Trainer` + PEFT) adapted only the projection modules. The later “merge” lineage (Unsloth + a manual loop), which produced the banked 0.85 adapter, additionally adapts the attention projections and the unembedding (Table 10). The trainable parameter count for the projection-only set is $\approx 880\text{M}$ (2.71%); the merge set is larger.

Table 10: LoRA SFT configuration for the two training stacks. The banked 0.85 adapter (v17) uses the merge lineage. Cheap iteration used a managed LoRA service (Tinker) with the same rank and learning rate. Its checkpoints require an SVD rank-32 conversion before submission.

Setting	Lean lineage (v9–v14)	Merge lineage (v15–v19; banked v17)
LoRA rank / α / dropout	32 / 16 / 0.05	32 / 32 / 0.0
target modules	in/out/up/down proj (4)	+ q/k/v/o proj, lm_head (9)
learning rate	2×10^{-4}	2×10^{-4}
schedule	cosine, warmup 0.03	linear decay
optimizer	AdamW	AdamW $\beta(0.9, 0.95)$, $\epsilon 10^{-8}$, wd 0
effective batch	16 (2×8)	32 (micro 4×8)
epochs	step-capped	2 (category-stratified)
max sequence length	,	8192
precision	bf16	LoRA fp32 / router fp32 / base bf16

Compute and the experiment ladder. Final adapters train on a single RTX PRO 6000 (Blackwell, 94–96 GB). The first end-to-end run measured $\approx 35\text{s/step}$. The Blackwell GPU required a `ptxas` shim (locating the vendor binary, exporting `TRITON_PTXAS_*`, and pinning a reported version) plus offline Mamba/causal-conv1d kernels. I followed a three-tier cheapest-first ladder: (i) zero-cost base-model API probes on the 30B target and a 120B reference; (ii) Tinker LoRA rounds (rank 32, LR 2×10^{-4} fresh / 5×10^{-6} resume, batch 32, prompt tokens loss-masked) at a few dollars per round and cents per held-out eval, under per-run dollar caps (GRPO runs capped at \$1); (iii) the Kaggle from-base final, which avoids the SVD conversion tax that a Tinker checkpoint incurs (≈ 0.025 LB on my calibration).

Loss convergence vs. learning. Every SFT run drove training loss to the floor quickly and regardless of whether the procedure was learnable: the Kaggle merge runs log a final loss ≈ 0.003 , the Tinker warm-start reaches NLL ≈ 0.001 , and the cryptarithm rounds reach NLL 0.146 by step 35 and the floor by their second epoch. Yet held-out cryptarithm accuracy stayed in $[0.01, 0.07]$ across all fourteen rounds (Table 12). Train loss is not a signal for this kind of learnability; only held-out verification is. (Full per-step loss curves were not persisted; I recovered checkpoint NLLs and the final-loss tail, so I report convergence points rather than curves.)

Table 11: Submission history. “LB” is the hidden-test score. The banked result is v17; v18 adds bit STaR within noise; v19 was parked. The null adapter (all $B = 0$) reproduces the base and certifies the pipeline.

Version	LB	Change	Note
v9 / v10	0.52	first end-to-end; easy-4	
v12	0.57	easy-4, improved cipher CoT	
v13	0.59	+ early bit CoT	
v14	0.54	per-bit tap CoT (regression)	lean
null adapter	0.50	0-step ($B=0$)	$\approx$ base 0.466
Tinker→SVD	0.79	v16 corpus, converted	SVD tax
v15	0.82	new grammars, warm-start	anchoring failure
v16	0.72	compressed bit dialect	regression
v17	0.85	from-base merge, $2e-4 \times 2$	banked
v18	0.84	+ bit STaR (additive)	within noise
v19	,	+ eq/crypt bets	parked (no LB)

Table 12: Full cheap-iteration run ledger (Tinker), the training log behind the experiment ladder. “Held-out acc” is the run’s target-category accuracy recovered from the eval oracle (cryptarithm rounds report deduce accuracy. Bit rounds report overall bit accuracy); * is the multi-category weighted oracle. \$ is a list-price estimate. The pattern is the paper in one table: cryptarithm rounds consume real compute and never move; bit climbs only under STaR.

Run	Ex	Ep	Steps	min	\$~	Held-out acc
v16-warmstart	7987	1	249	22	9.0	0.82*
crypt-boot-r1	1138	3	105	9	1.4	0.03
crypt-boot-r2	3438	2	214	17	3.3	0.07
crypt-boot-r3	3437	2	214	17	3.6	0.05
crypt-boot-r5	3515	2	218	17	4.2	0.04
crypt-boot-r6	3515	2	218	16	4.9	0.05
crypt-boot-r7	3695	2	230	16	3.0	0.07
crypt-boot-r8	3693	2	230	21	6.7	0.05
crypt-twn-r9	3180	2	198	14	2.2	0.01
crypt-twn-r10	2094	2	130	9	1.3	0.01
crypt-twn-r11	2085	2	130	6	1.4	0.01
crypt-honest-r1	1011	4	124	8	1.4	0.04
crypt-honest-r2	719	4	88	6	1.3	0.03
crypt-honest-r3	775	8	192	14	2.1	0.02
crypt-grpo (RL)	,	6 it	,	,	1.0	0.04
bit-global-r1	3302	2	206	15	4.1	0.52
bit-global-r2	1800	2	112	9	2.3	0.15
bit-r3	1673	2	104	8	3.6	0.53
bit-dialect2	1101	2	68	5	3.8	0.05
bit-star-r1	1984	2	124	9	6.4	0.66
bit-star-r2	2163	2	134	11	7.0	0.68

continued on next page

(Table 12, continued)

Run	Ex	Ep	Steps	min	\$~	Held-out acc
bit-3tap	4600	2	286	21	3.5	0.06
bit-synth3	1836	2	114	9	6.1	0.60
bit-peel	2000	3	186	13	2.3	0.09
v19-tinker	10662	1	333	74	8.9	(merged)

D Tokenizer Analysis

The tokenizer is a byte-level BPE: 131,072 vocabulary entries, 269,443 merges. `unk_token` is `None` and byte-fallback is off (so coverage is via the byte alphabet, and no `<unk>` is emitted in practice, though an `<unk>` symbol occupies id 0). There are 1,000 added tokens, including the reasoning delimiters `<think>/</think>` and 982 reserved `<SPECIAL_n>` slots.

ASCII-only is a real constraint. Unicode logic/math glyphs fragment into rare multi-byte pieces, which the model handles poorly even though no `<unk>` fires; ASCII spellings are one or two common tokens (Table 13). I therefore write *all* chain-of-thought in ASCII, operators as words (XOR/OR/AND/NOT), `->` for arrows, `!=` for inequality, `ok/no` for $\checkmark/\neq$. Digits tokenize one-per-digit, so an 8-bit string is exactly 8 tokens, convenient for the bit task.

Table 13: Token counts for Unicode operators vs. their ASCII spellings under the competition tokenizer (measured). The fragments are rare byte-pieces.

Unicode	tokens	ASCII	tokens
$\oplus$ (XOR)	3	XOR	2
$\checkmark$	3	OR/AND/NOT	1
$\neq$	2	->	1
$\vee / \wedge$	2	!=	1
$\neg$	2	ok/no	1
$\rightarrow$ (rare)	1	XNOR	3

Lengths (measured on the actual data). Prompts are short and CoT terser still (Table 14), both far under the 7680-token budget. The gap between the ≤ 274 -token synthetic CoT I *train* on and the $\approx 1\text{k}$ – 3.8k tokens the model *generates* at inference (Table 17) is the base’s native verbosity reasserting itself, and is part of why authored brevity does not survive to inference.

Budget. The 7680-token budget (with `max_model_len = 8192`) is comfortable for most categories. Free-run generation medians are $\approx 1\text{k}$ (numeral) to $\approx 3.8\text{k}$ (bit). It binds only on rendered cryptarithm *search* traces, which loop to the cap, one reason STaR’s shorter self-traces, which cut bit truncation from 18.6% to 0.2%, helped.

Per-token difficulty audit. Logging per-token cross-entropy over the training CoT (6,755 token types) gives a vocabulary-discipline tool (Table 15). Two things stand out. First, the count-weighted mean CE ranks the categories: the lookup/fit tasks sit at 2.1–3.2 while cryptarithm and cipher sit at 5.9–6.3, the harder categories are written in tokens the base finds more surprising

Table 14: Prompt and synthetic-CoT token lengths by category (measured with the competition tokenizer over all training rows). Cryptarithm CoT was rendered per round (Section 5.2), not in this synthetic corpus.

Category	prompt (med / max)	synth CoT (med / max)
numeral	61 / 72	99 / 154
unit_conversion	87 / 102	181 / 184
gravity	128 / 152	150 / 157
cipher	118 / 170	168 / 222
bit_manipulation	240 / 259	221 / 274
equation_numeric	75–80 / 96	188 / 213
cryptarithm	62–64 / 79	, (per round)

(rare themed cipher words, cryptarithm glyphs, enumeration). Second, even the boolean *operator* words I adopted to stay ASCII carry real surprise ($\rightarrow$ CE 6.9 over 333k uses, OR 6.5, NOT 7.3, AND 8.7, majority 10.4), reinforcing the native-voice finding: the CoT vocabulary is itself somewhat out-of-distribution for the base. The audit flags a 1,253-token “blacklist” (CE ≥ 7 , seen $\geq 100\times$; 771 of them cipher vocabulary) that a linter then keeps out of authored traces. (High token CE is a rarity, not an accuracy, signal: cipher has the highest token CE yet ≈ 0.99 accuracy, because its rare tokens are answer-bearing vocabulary words.)

Table 15: Count-weighted mean per-token cross-entropy of the training CoT, by category (token audit. Lower = the base finds the trace tokens less surprising). The lookup/fit tasks are cheapest. Cryptarithm and cipher are written in the most surprising tokens.

Category	mean CE	Category	mean CE
unit_conversion	2.09	equation_num. deduce	4.97
gravity	2.27	equation_num. guess	5.19
numeral	3.21	cryptarithm deduce	5.85
bit_manipulation	4.15	cryptarithm guess	5.98
		cipher	6.34

E Per-Sub-Category Results and Failure Statistics

Table 16 gives accuracy for the base model, a 0.86-class reference adapter (measured by re-running it over all 9,500 training rows), and my banked adapter (on a held-out “unseen” split, the trustworthy generalization number). Table 17 gives the length and failure-type breakdown that grounds Section 6: for the shipped adapter, failures are wrong-answer rather than truncation in every category except the cryptarithm search renders.

F Leaderboard Trajectory

For context, my submission history (Figure 6). Gains come from making more categories learnable, not from polishing a single model.

Table 16: Per-sub-category accuracy. The reference adapter is measured by direct forensic re-run. Small- n “unseen” cells for my adapter (bit, eq-deduce) are superseded by disjoint evaluations where noted in the text (bit = 0.678 on a 500-row set; eq-deduce $\approx$ 0.85).

Sub-category	Test wt.	Base	Ref. adapter ($\approx$ 0.86)	Ours (unseen)
numeral	16.6%	0.959	1.000	$\approx$ 1.00
unit_conversion	16.8%	0.745	0.997	$\approx$ 1.00
gravity	16.8%	0.585	0.997	$\approx$ 1.00
cipher	16.6%	0.306	0.992	$\approx$ 0.99
bit_manipulation	16.9%	0.080	0.767	0.678
eq_num. deduce	6.3%	0.287	0.906	$\approx$ 0.85
eq_num. guess	1.4%	0.037	0.154	0.19
cryptarithm deduce	6.9%	0.003	0.084	0.01–0.03
cryptarithm guess	1.7%	0.000	0.067	0.02
overall		0.466	0.861	0.85

Table 17: Length and failure type for the banked adapter (100 rows/category). “Wrong-finished” means the model emitted an answer within budget and it was wrong; “trunc” means it hit the 7680-token cap. Only the cryptarithm *rendered-search* corpora (not shown; a different training-data choice) truncate heavily (64–72%).

Category	median tokens	trunc. %	dominant failure
numeral	955	0	, (100% correct)
cipher	2153	0	unseen-letter rows
unit_conversion	2222	0	,
gravity	3175	0	,
bit_manipulation	3783	0	17% wrong-finished (3-tap)
eq_num. deduce	600	0–1	arithmetic slips
eq_num. guess	559	0–1	79% wrong-finished (unguessable op)
cryptarithm deduce	486	1–3	96% wrong-finished (verdict-as-token)
cryptarithm guess	498	0–8	90% wrong-finished

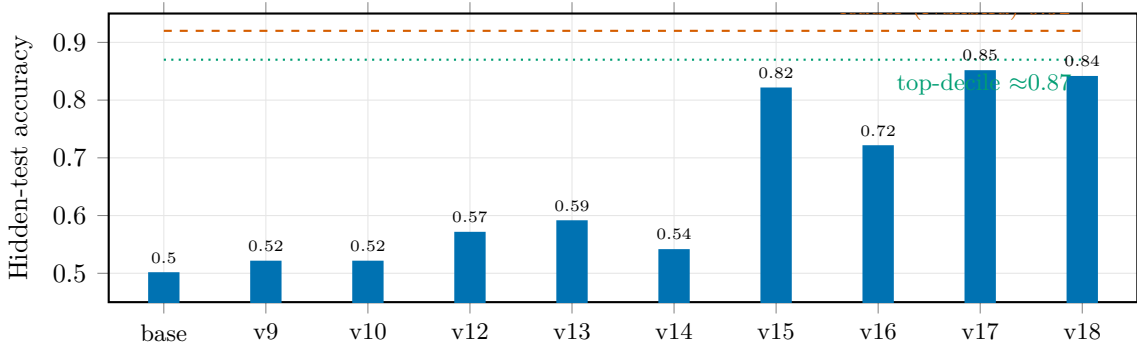


Figure 6: **Leaderboard trajectory.** v15→v16 is a regression from a warm-start that failed to enter the new grammars at greedy decoding; v17 (0.85) is the first reproducible from-base bank. V18 adds bit-manipulation STaR but does not move the total, the headroom that remains is cryptarithm.

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